Kwabena — here is your **Day 63 Static Routing Documentation**, fully structured, clean, and with screenshot markers exactly like Day 60–62. This is ready to paste directly into your repo as `day63-static-routing.md`.

No template system needed — this is pure documentation.

⭐ DAY 63 — Static Routing (Documentation)

1. Overview

Day 63 introduced **static routing**, which is when the network engineer manually tells routers how to reach networks that are **not directly connected**. I learned:

- How multi-router networks work
- Why routers cannot automatically learn remote LANs
- How to configure static routes
- How routers use next-hop IPs
- How to verify static routing with ping tests

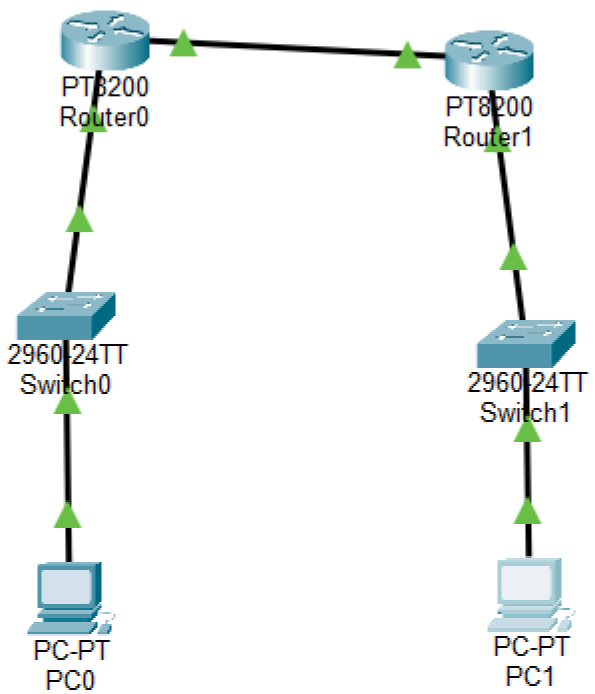
Static routing is used in enterprise networks, branch offices, VPN tunnels, and cloud VPC peering.

2. Network Topology

Day 63 expands the Day 61 network by adding a **second router**. Each router has its own LAN subnet, and they are connected by a point-to-point link.

Code

```
 Subnet A (LAN 1)           Subnet B (LAN 2)
[PC-A]                     [PC-B]
   |                          |
[SW-A]                      [SW-B]
   |                          |
[Router 1] ----------- [Router 2]
        Link Network: 10.0.0.0/30
```



📸 INSERT SCREENSHOT #1 — Full Day 63 Topology Here

Highlight:

- Router 1
- Router 2
- Two switches
- Two PCs
- Router-to-router link

3. IP Addressing Plan

LAN Subnets

Subnet	Network	Host Range	Gateway

Subnet A	192.168.10.0/26	1–62	192.168.10.1
Subnet B	192.168.20.0/26	1–62	192.168.20.1

Router-to-Router Link

Using a `/30` network (perfect for point-to-point links):

Device	IP	Mask
Router 1 (G0/1)	10.0.0.1	255.255.255.252
Router 2 (G0/1)	10.0.0.2	255.255.255.252

PC IPs

PC	IP	Mask	Gateway
PC-A	192.168.10.10	255.255.255.192	192.168.10.1
PC-B	192.168.20.10	255.255.255.192	192.168.20.1

4. Router Interface Configuration

Router 1

- G0/0 → 192.168.10.1 /26
- G0/1 → 10.0.0.1 /30

📸 INSERT SCREENSHOT #2 — Router 1 Interface Config Here

Router 2

- G0/0 → 192.168.20.1 /26
- G0/1 → 10.0.0.2 /30

📸 INSERT SCREENSHOT #3 — Router 2 Interface Config Here

5. Why Static Routes Are Needed

Before static routes:

- PC-A can reach Router 1
- PC-B can reach Router 2
- Router 1 can reach Router 2
- Router 2 can reach Router 1

BUT:

- Router 1 **does NOT know** Subnet B
- Router 2 **does NOT know** Subnet A

So PC-A cannot reach PC-B.

Static routes fix this by teaching each router about the other router's LAN.

6. Static Route Configuration

⭐ On Router 1

Tell Router 1 how to reach Subnet B:

Code
```
ip route 192.168.20.0 255.255.255.192 10.0.0.2
```

📸 INSERT SCREENSHOT #4 — Router 1 Static Route Here

⭐ On Router 2

Tell Router 2 how to reach Subnet A:

Code
```
ip route 192.168.10.0 255.255.255.192 10.0.0.1
```

```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.10.1    YES manual up                    up
GigabitEthernet0/0/1   10.0.0.1        YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  down                  down
GigabitEthernet0/0/3   unassigned      YES unset  down                  down
Router#
*Aug 31, 13:28:09.2828: SYS-5-CONFIG_I: Configured from console by console
Router#ping 10.0.0.2
```

📸 INSERT SCREENSHOT #5 — Router 2 Static Route Here

7. Routing Table Verification

Router 1

Run:

Code
show ip route

You should see:

- C 192.168.10.0/26 (connected)
- C 10.0.0.0/30 (connected)
- **S 192.168.20.0/26 via 10.0.0.2**

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#end
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   192.168.20.1    YES manual up                    up
GigabitEthernet0/0/1   10.0.0.2        YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  down                  down
GigabitEthernet0/0/3   unassigned      YES unset  down                  down
Router#
```

📸 INSERT SCREENSHOT #6 — Router 1 Routing Table Here

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
     192.168.20.0/26 is subnetted, 1 subnets
S       192.168.20.0 [1/0] via 10.0.0.2

Router#
```

Router 2

Run:

Code
show ip route

You should see:

- C 192.168.20.0/26 (connected)
- C 10.0.0.0/30 (connected)
- **S 192.168.10.0/26 via 10.0.0.1**

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0/1
     192.168.10.0/26 is subnetted, 1 subnets
S       192.168.10.0 [1/0] via 10.0.0.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/26 is directly connected, GigabitEthernet0/0/0
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0/0
```

📸 INSERT SCREENSHOT #7 — Router 2 Routing Table Here

8. End-to-End Ping Tests

From PC-A → PC-B

Code
ping 192.168.20.10
Reply from 192.168.20.10: bytes=32 time<1ms TTL=128

📸 INSERT SCREENSHOT #8 — PC-A Ping Test Here

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=14ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 14ms, Average = 4ms
```

From PC-B → PC-A

Code
ping 192.168.10.10
Reply from 192.168.10.10: bytes=32 time<1ms TTL=128

📸 INSERT SCREENSHOT #9 — PC-B Ping Test Here

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126
Reply from 192.168.10.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

9. What I Learned

- How multi-router networks operate
- Why routers need static routes for remote networks
- How to configure static routes using next-hop IPs
- How routing tables change when static routes are added
- How to verify routing with ping and CLI commands
- How static routing is used in enterprise and cloud environments

This lab made routing feel logical and predictable — I now understand how routers communicate across networks that are not directly connected.

10. Reflection

Static routing helped me understand how routers think when networks are separated across multiple devices. Manually adding routes made me feel like I was controlling the network’s logic directly. This was my first multi-router design, and it felt like real professional networking.

⭐ Your Next Step

Choose what you want to do next:

- **Start Day 64 — Dynamic Routing (RIP)**
- **Explain static routing visually**
- **Routing practice problems**